Online Constrained Control of Storage Systems via Simplex Disturbance-Action Policies

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***Abstract*—We study online control of a scalar storage system with nonnegative adversarial resource arrivals and state-dependent action constraints. The adversarial, time-varying cost depends on both the state and the action. At each time, the controller selects a feasible action before the current resource arrival and cost are revealed. To handle the resulting coupling between feasibility and online learning, we introduce *Simplex Disturbance-Action Control* (SDAC), a policy class specified by H nonnegative parameters whose sum is at most one. The key feature of SDAC is that every fixed policy in the class is feasible by construction. Our online SDAC controller updates these parameters using entropic online mirror descent while preserving feasibility along the time-varying trajectory. We prove a regret bound of $O\Big(\sqrt{T\log(H+1)}\Big)$ relative to the best fixed SDAC policy. We also derive a minimax lower bound showing that the dependence on T is tight. We then show that SDAC geometrically approximates a decreasing-allocation policy class that spreads each arrival over time according to a common decreasing allocation sequence, a structure that arises in stochastic energy-harvesting control. With $H = \Theta(\log T)$, the resulting regret against the decreasing-allocation class is $O\big(\sqrt{T\log\log T}\big)$. The framework applies to energy-harvesting batteries and other storage-constrained resource systems.**

***Index Terms*—Online control, disturbance-action policies, storage systems, energy harvesting, regret minimization, state-dependent action constraints.**

I. Introduction

A. Motivation and Problem Setting

MANY resource-allocation systems have a storage state that changes over time and limits the control action at each time. An action that lowers the current cost may leave less resource for future decisions, while conserving the resource may increase the current cost. The controller must balance these effects while satisfying this constraint at every step.

We study this problem for a scalar storage system that loses a fixed fraction of its stored resource each period. At each time, the controller observes the current state and all past arrivals and cost functions, chooses a control action, and then observes the current arrival and cost function. The arrival and cost sequences may be chosen adversarially, so we do not assume a known probability model. Our goal is to construct an online SDAC controller whose cumulative cost is close to that of the best fixed policy, chosen in hindsight, from a specified benchmark class.

One difficulty is that the current action affects the next state. In online control, *Disturbance-Action Control* (DAC) parameterizes actions using a linear filter of past disturbances, thereby yielding a tractable algorithm with bounded regret. In our storage model, however, some DAC policies can produce negative actions or actions larger than the available resource.

We address this issue by introducing *Simplex Disturbance-Action Control* (SDAC). SDAC restricts the filter parameters to a subprobability simplex and includes powers of the retention coefficient in the filter. These choices ensure that every fixed SDAC policy satisfies the action constraint. The state and action components of each fixed-policy trajectory are also affine in the policy parameter. Thus, a convex state–action cost is convex as a function of that parameter. This structure permits the parameter to be updated using online convex optimization.

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B. Model and Information Pattern

The storage state $x_t \geq 0$ evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t,$$

where $\alpha \in (0,1)$ is a fixed retention coefficient, u_t is the control action, and $w_t \in [0,1]$ is a nonnegative resource arrival. The action must satisfy

$$0 \leq u_t \leq \alpha x_t.$$

Thus, the controller cannot use more than the amount remaining after storage loss.

At time t, the controller chooses u_t using the current state and the information available through time $t-1$. The arrival w_t and the convex cost function c_t are revealed after the action is selected, and the controller incurs the cost $c_t(x_t, u_t)$. Both the online trajectory and every fixed-policy trajectory used as a benchmark must satisfy the same dynamics and action constraint. The formal model and assumptions are given in Section II.

Energy-harvesting battery management is our main application. In this case, x_t is the battery state of charge, w_t is the harvested energy, u_t is the energy used, and $1-\alpha$ is the fraction of stored energy lost during one period. Similar dynamics arise in perishable inventory systems, water reservoirs with storage loss, and queues with abandonment. In many standard queueing models, $\alpha = 1$; that case is outside our main results and is discussed in Section VII.

C. Contributions

Our contributions are as follows:

- **Policy class.** We introduce SDAC, a finite-memory policy class parameterized by a subprobability simplex. We

prove that every fixed SDAC policy is feasible for every arrival sequence with entries in $[0, 1]$.
We further show that SDAC approximates the decreasing-allocation policy class defined in Section III-C. This class allocates each arrival over future periods according to the same decreasing sequence. It contains every feasible proportional state-feedback policy $u_t = kx_t$, with $k \in [0, \alpha]$, and the approximation error decreases geometrically with the memory horizon H.

- **Online control and regret.** We develop an online SDAC controller that updates its parameter using entropic online mirror descent (OMD) over the subprobability simplex. We prove that the controller remains feasible and achieves

$$O\Big(\sqrt{T\log(H+1)}\Big)$$

regret against the best fixed SDAC policy. For $H = \Theta(\log T)$, the regret against the decreasing-allocation class is $O\big(\sqrt{T\log\log T}\big)$. The same bound holds against the best proportional state-feedback policy.
- **Minimax lower bound.** We prove a lower bound of order $\sqrt{T}$ for the expected regret of every causal feasible controller, including randomized controllers; see Theorem 2. When T is at least on the order of $(1-\alpha)^{-1}$, the lower bound scales as $\Omega((1-\alpha)^{-1}\sqrt{T})$ near $\alpha = 1$. The upper and lower bounds differ by a factor of $(1-\alpha)^{-1/2}$ in their dependence on $1-\alpha$. Closing this gap is left for future work.

D. Related Work

a) Online control: Online control is the main basis for this work. Agarwal et al. use DAC policies to obtain $\widetilde{O}(\sqrt{T})$ regret against strongly stable linear controllers [1]. More closely related to our simplex parameterization, Golowich et al. adapt DAC to population dynamics and search over simplex-structured DAC policy parameters [2]. Their state lies in a probability simplex, and their feasible action set is independent of the current state. Here, the constraint $0 \le u_t \le \alpha x_t$ couples each action to the available resource.

Li et al. consider affine constraints on the state and action [3]. Their action constraint is independent of the current state. They enforce feasibility by restricting and projecting their policy updates onto a tightened parameter set. In our work, the action produced by every fixed SDAC policy is feasible by construction. When the parameter changes online, we clip the nominal action to the current admissible interval as in (30).

Liu et al. consider general convex constraint functions [4]. For their adversarial constraints, the guarantees bound cumulative violation and may permit violations at individual times. Such violations are not permitted here, since (2) must hold at every time.

Kumar et al. introduce online convex optimization with unbounded memory [5]. For their discounted infinite-memory model, the lower bound scales as $\sqrt{T}$ divided by one minus the discount factor. This parallels our $\Omega((1-\alpha)^{-1}\sqrt{T})$ lower bound beyond the mixing time, although ours applies directly to causal feasible controllers under the state-dependent action constraint.

b) Energy harvesting: Fixed-fraction and related allocation policies for stochastic energy arrivals motivate our *decreasing-allocation policy class*. Shaviv and Özgür [6] allocate a fixed fraction of the available energy under i.i.d. arrivals. Arafa et al. [7] extend fixed-fraction policies to general concave utilities under i.i.d. arrivals and, for Bernoulli arrivals, characterize a non-increasing allocation sequence following a full recharge. Zibaeenejad and Chen [8] derive an optimal decreasing allocation sequence under Bernoulli arrivals when the controller observes a fixed look-ahead window. Together, these works motivate allocating each arrival over future periods according to a decreasing sequence. Our allocation begins one period after the arrival because the controller acts before observing the current arrival.

Yu and Neely [9] study expected utility maximization under i.i.d. system states, while Asgari and Neely [10] obtain an $O(\sqrt{T})$ regret–battery-capacity tradeoff for arbitrary loss sequences under i.i.d. energy arrivals. Neither objective depends directly on the battery state. Yu and Neely optimize expected time-average utility, whereas Asgari and Neely measure regret against a fixed action. Lin et al. [11] allow arbitrary energy arrivals and time-varying age-of-information objectives. They observe the current arrival before allocation and compare with an unrestricted offline optimum through a competitive ratio. In our setting, the controller acts before observing the current arrival and cost, the arrival and cost sequences may be chosen adversarially, the cost depends on the storage state and action, and the benchmark is a fixed SDAC policy chosen in hindsight.

c) Queueing and network optimization: Classical queueing control studies throughput and queue stability through dynamic service and scheduling decisions [12]–[14], while stochastic network optimization studies tradeoffs between time-average performance and average queue backlog [15]. Our performance criterion is finite-horizon regret against the best fixed SDAC policy in hindsight.

For their time-invariant class, Khojastepour and Sabharwal [16] represent schedulers satisfying a maximum-delay constraint as fixed linear filters of the current and past arrivals, with nonnegative coefficients summing to one rather than at most one as in SDAC. Under i.i.d. arrivals, they identify the uniform moving-average filter as the minimizer of long-run average transmit power over a Gaussian channel. The connection is limited to the simplex-like filter structure: they solve a stochastic average-power problem with a fixed cost, whereas our algorithm updates the SDAC parameters online under adversarial arrival and cost sequences with general convex state–action costs.

d) Inventory and reservoir management: Yang et al. [17] study online procurement under inventory constraints and obtain optimal competitive ratios relative to an offline optimum. They observe the current demand and price before acting, whereas our controller acts before the current arrival and cost are revealed. Hihat and Fermanian [18] learn a base-stock policy for more general inventory dynamics but do not prove a regret bound for their method. For our scalar linear model, we prove feasibility, approximation, and sublinear regret guarantees. Classical stochastic reservoir models study storage under random inflows and fixed release rules [19], with

an emphasis on steady-state behavior rather than finite-horizon online regret.

II. Model and Regret Benchmark

For every positive integer n, we write $[n] \triangleq \{1, \dots, n\}$. For every $z \in \mathbb{R}$, we define $[z]_+ \triangleq \max\{z, 0\}$. For $\mathbf{v} \in \mathbb{R}^n$, the i-th coordinate is denoted by v_i. If the vector already carries a subscript, the coordinate is appended after a comma: the i-th coordinate of $\mathbf{v}_t$ is $v_{t,i}$. We write $\mathbf{1}$ for the all-ones vector, with its dimension clear from context, and $\langle \cdot, \cdot \rangle$ for the Euclidean inner product. We use $\log$ for the natural logarithm.

Fix a positive integer T and a known retention coefficient $\alpha \in (0, 1)$. The state $x_t \in \mathbb{R}_{\geq 0}$ is the resource level at the beginning of time t. The system starts empty:

$$x_1 = 0. \tag{1}$$

At each time $t \in [T]$, the controller chooses an action $u_t \in \mathbb{R}_{\geq 0}$ satisfying

$$0 \leq u_t \leq \alpha x_t. \tag{2}$$

The system then receives an arrival $w_t \in [0, 1]$ and evolves according to

$$x_{t+1} = \alpha x_t - u_t + w_t. \tag{3}$$

Here, $1 - \alpha$ is the fraction of the stored resource lost during one period. The constraint (2) ensures that the action does not exceed the amount remaining after this loss.

In the online-control literature, w_t is commonly called a disturbance. Here, we call it an *arrival* because it is nonnegative, while retaining the standard names Disturbance-Action Control. The nonnegativity of w_t is important for both the storage interpretation and the feasibility properties of SDAC.

At time t, the controller chooses u_t before observing w_t or c_t. After the action is selected, the arrival and the entire cost function are revealed, the controller incurs $c_t(x_t, u_t)$, and the state is updated according to (3).

For every feasible trajectory, (3) and $w_t \in [0, 1]$ give

$$0 \leq x_{t+1} \leq \alpha x_t + 1.$$

Since $x_1 = 0$, induction gives

$$0 \leq x_t \leq \frac{1 - \alpha^{t-1}}{1 - \alpha} \leq \frac{1}{1 - \alpha}, \qquad t \in [T + 1].$$

Together with (2), this bound places every feasible state–action pair in the compact set

$$\mathcal{D}_\alpha \triangleq \left\{ (x, u) : 0 \leq x \leq \frac{1}{1 - \alpha}, 0 \leq u \leq \alpha x \right\}. \tag{4}$$

Remark 1 (Storage capacity). A storage capacity of $1/(1-\alpha)$ suffices to avoid overflow.

Each revealed cost function has the form

$$c_t : \mathbb{R}^2_{\geq 0} \to \mathbb{R}$$

and is convex in (x, u). We also assume that each c_t satisfies the following Lipschitz condition on $\mathcal{D}_\alpha$: there are constants $L_x, L_u > 0$, such that

$$|c_t(x, u) - c_t(x', u')| \leq L_x |x - x'| + L_u |u - u'| \tag{5}$$

for all $(x, u), (x', u') \in \mathcal{D}_\alpha$. We also assume access, at every $(x, u) \in \mathcal{D}_\alpha$, to a cost subgradient

$$\boldsymbol{\zeta}_t(x, u) = \big(\zeta_t^x(x, u), \zeta_t^u(x, u)\big) \in \partial c_t(x, u)$$

satisfying

$$|\zeta_t^x(x, u)| \leq L_x, \qquad |\zeta_t^u(x, u)| \leq L_u.$$

These subgradient bounds hold, for example, when (5) extends to an open neighborhood of $\mathcal{D}_\alpha$; see [20, Thm. 9.13].

We use x_t and u_t without parameter arguments for the trajectory generated by the online controller. Its cumulative cost is

$$J_T^{\mathrm{on}} \triangleq \sum_{t=1}^{T} c_t(x_t, u_t). \tag{6}$$

For a fixed positive integer H, let $\pi_{\mathbf{m}}$ denote the fixed SDAC policy indexed by $\mathbf{m} \in \mathcal{M}_H$, where $\mathcal{M}_H$ is defined in Section III. The *fixed-policy trajectory associated with* $\mathbf{m}$ is the unique state–action sequence $\{(x_t(\mathbf{m}), u_t(\mathbf{m}))\}_{t=1}^T$ obtained by applying $\pi_{\mathbf{m}}$ with $\mathbf{m}$ held constant at every time. Define the cumulative cost of this fixed-policy trajectory by

$$J_T(\mathbf{m}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big).$$

The regret against the best fixed SDAC policy is defined by

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \triangleq J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_H} J_T(\mathbf{m}). \tag{7}$$

Our objective is to satisfy (2) at every time and achieve sublinear regret.

For the other comparator classes considered below, $x_t(k)$ and $u_t(k)$ denote the state and action induced by a feasible proportional state-feedback policy with gain $k \in [0, \alpha]$, while $x_t(\mathbf{q})$ and $u_t(\mathbf{q})$ denote those induced by a decreasing allocation sequence $\mathbf{q}$.

III. Simplex Disturbance-Action Control (SDAC)

A. From Disturbance-Action Control to SDAC

DAC uses a finite linear filter of past disturbances. Under standard conditions, this parameterization yields convex surrogate costs and efficient online updates. In our scalar storage model, DAC has the following form:

$$u_t = \sum_{i=1}^{H} \theta_i w_{t-i}, \tag{8}$$

where H is a positive integer (the memory horizon) and $\boldsymbol{\theta} = (\theta_1, \dots, \theta_H)^\top \in \mathbb{R}^H$ is a linear filter.

For our constrained model, an unrestricted filter does not ensure feasibility: its action can exceed αx_t or become negative. We therefore introduce *Simplex Disturbance-Action Control* (SDAC), which uses a simplex-constrained filter tailored to the storage dynamics. This restriction ensures that every fixed SDAC policy is feasible while preserving the convexity of the surrogate costs and enabling us to achieve a tractable online algorithm.

B. Policy Definition and Structural Properties

Fix a memory horizon $H \in \mathbb{N}_{>0}$ and define the subprobability simplex

$$\mathcal{M}_H \triangleq \left\{\mathbf{m} \in \mathbb{R}^H_{\geq 0} : \|\mathbf{m}\|_1 \leq 1\right\}. \tag{9}$$

With the convention $w_\tau \equiv 0$ for $\tau \leq 0$, each $\mathbf{m} \in \mathcal{M}_H$ defines an SDAC policy $\pi_{\mathbf{m}}$ through

$$u_t(\mathbf{m}) \triangleq \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = \langle \boldsymbol{\phi}^u_t, \mathbf{m} \rangle, \tag{10}$$

where

$$\boldsymbol{\phi}^u_t \triangleq \left(\alpha w_{t-1}, \alpha^2 w_{t-2}, \ldots, \alpha^H w_{t-H}\right)^\top \in \mathbb{R}^H. \tag{11}$$

The resulting SDAC policy class is

$$\Pi^{\mathrm{SDAC}}_H \triangleq \{\pi_{\mathbf{m}} : \mathbf{m} \in \mathcal{M}_H\}. \tag{12}$$

We next establish two structural properties of this policy class. Every $\pi_{\mathbf{m}}$ satisfies the feasibility constraint (2), and its induced cost is convex and Lipschitz in $\mathbf{m}$. Proofs omitted from this section are collected in Appendix A.

Lemma 1 (State decomposition under SDAC)**.** *Fix a positive integer H and $\mathbf{m} \in \mathcal{M}_H$. For each $t \in [T]$, define*

$$r_t(i) \triangleq \sum_{j=1}^{i} \alpha^{j-1} w_{t-j}, \quad i \in [H], \qquad r_t \triangleq \sum_{j=1}^{t-1} \alpha^{j-1} w_{t-j},$$

where $r_1 = 0$. Then the state of a fixed-policy induced by $\pi_{\mathbf{m}}$ satisfies, for every $t \in [T]$,

$$x_t(\mathbf{m}) = \sum_{i=1}^{H} m_i r_t(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_t. \tag{13}$$

Equivalently, define

$$\boldsymbol{\phi}^x_t \triangleq \left(r_t(1) - r_t, \ldots, r_t(H) - r_t\right)^\top.$$

Then

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}^x_t, \mathbf{m} \rangle.$$

Moreover,

$$0 \leq r_t(i) \leq r_t \leq \frac{1}{1-\alpha},$$

for every $t \in [T]$ and $i \in [H]$. Consequently, $\mathbf{m} \mapsto x_t(\mathbf{m})$ is affine for every $t \in [T]$.

Lemma 2 (Feasibility of SDAC policies)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the trajectory induced by $\pi_{\mathbf{m}}$ satisfies the action constraint* (2)*. Explicitly,*

$$0 \leq u_t(\mathbf{m}) \leq \alpha x_t(\mathbf{m}) \qquad \text{for all } t \in [T]. \tag{14}$$

Proof. Because m_i, α, and w_{t-i} are nonnegative,

$$u_t(\mathbf{m}) = \sum_{i=1}^{H} m_i \alpha^i w_{t-i} \geq 0.$$

For every $i \in \{1, \ldots, H\}$,

$$\alpha r_t(i) = \sum_{j=1}^{i} \alpha^j w_{t-j} \geq \alpha^i w_{t-i}. \tag{15}$$

Because $\mathbf{m} \in \mathcal{M}_H$ and $r_t \geq 0$ by Lemma 1,

$$\begin{aligned} \alpha x_t(\mathbf{m}) &= \sum_{i=1}^{H} m_i \alpha r_t(i) + \alpha\big(1 - \sum_{i=1}^{H} m_i\big) r_t \\ &\geq \sum_{i=1}^{H} m_i \alpha r_t(i) \\ &\geq \sum_{i=1}^{H} m_i \alpha^i w_{t-i} = u_t(\mathbf{m}), \end{aligned} \tag{16}$$

where the second inequality uses (15). □

For any realized arrival history, both components of the fixed-policy trajectory are affine in $\mathbf{m}$. The following lemma records the resulting convexity and the constants used by the online update.

Lemma 3 (Convexity and Lipschitz continuity of surrogate cost)**.** *Define*

$$\ell_t(\mathbf{m}) \triangleq c_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big), \qquad \mathbf{m} \in \mathcal{M}_H,$$

and

$$G_\alpha \triangleq \alpha\left(\frac{L_x}{1-\alpha} + L_u\right). \tag{17}$$

Then ℓ_t is convex on $\mathcal{M}_H$, and, for all $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| \leq G_\alpha \|\mathbf{m} - \mathbf{m}'\|_1.$$

Moreover, at every $\mathbf{m} \in \mathcal{M}_H$, one can select $\mathbf{g}_t \in \partial \ell_t(\mathbf{m})$ such that $\|\mathbf{g}_t\|_\infty \leq G_\alpha$.

For the online update, let (ζ^x_t, ζ^u_t) be the bounded cost subgradient at the state–action pair of the fixed-policy trajectory associated with $\mathbf{m}$ and use the explicit choice

$$\mathbf{g}_t = \zeta^x_t \boldsymbol{\phi}^x_t + \zeta^u_t \boldsymbol{\phi}^u_t.$$

This choice belongs to $\partial \ell_t(\mathbf{m})$ and satisfies $\|\mathbf{g}_t\|_\infty \leq G_\alpha$, as stated in Lemma 3.

C. Approximation of Decreasing-Allocation Policies

We next show that finite-memory SDAC policies approximate a class of policies that spread each resource arrival over all subsequent periods. Given a fixed decreasing sequence $(q_i)_{i \geq 1}$, an arrival w_τ contributes $q_i w_\tau$ to the expenditure at time $\tau + i$. The one-period delay reflects that w_τ is observed only after u_τ is chosen. Such allocation-sequence policies have been studied for Bernoulli recharges [7], [8].

Define the set of feasible decreasing allocation sequences

$$\mathcal{Q}_\alpha \triangleq \left\{\mathbf{q} = (q_i)_{i \geq 1} : q_1 \geq q_2 \geq \cdots \geq 0, \quad \sum_{i=1}^{\infty} \frac{q_i}{\alpha^i} \leq 1\right\}. \tag{18}$$

For $\mathbf{q} \in \mathcal{Q}_\alpha$, define

$$\begin{aligned} u_t(\mathbf{q}) &\triangleq \sum_{i=1}^{t-1} q_i w_{t-i}, \\ x_{t+1}(\mathbf{q}) &= \alpha x_t(\mathbf{q}) - u_t(\mathbf{q}) + w_t, \qquad x_1(\mathbf{q}) = 0. \end{aligned} \tag{19}$$

Here, $q_i w_s$ is the amount from arrival w_s allocated i periods later. The weighted budget in (18) accounts for leakage:

reserving q_i units for expenditure i periods after an arrival requires q_i/α^i units at the arrival time. Let

$$\Pi_\alpha^{\mathrm{alloc}} \triangleq \{\pi_{\mathbf{q}} : \mathbf{q} \in \mathcal{Q}_\alpha\}.$$

Lemma 4 (Finite-memory approximation of decreasing-allocation policies)**.** *Every policy in $\Pi_\alpha^{\mathrm{alloc}}$ is feasible. Fix $\mathbf{q} \in \mathcal{Q}_\alpha$ and $H \geq 1$, and define*

$$m_{H,i}(\mathbf{q}) \triangleq \frac{q_i}{\alpha^i}, \qquad i \in [H]. \tag{20}$$

Then $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. With

$$\mathrm{Tail}_H(\mathbf{q}) \triangleq \sum_{i=H+1}^{\infty} q_i,$$

the corresponding trajectories satisfy, for every arrival sequence in $[0,1]$,

$$0 \leq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) \leq \mathrm{Tail}_H(\mathbf{q}), \tag{21}$$

$$0 \leq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}) \leq \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha}. \tag{22}$$

If each c_t satisfies (5)*, then*

$$\begin{aligned}\left|\sum_{t=1}^{T}\left[c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\right]\right| \\ \leq \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q})T \\ \leq \left(L_u + \frac{L_x}{1-\alpha}\right)\alpha^{H+1}T.\end{aligned} \tag{23}$$

Corollary 1 (Proportional state feedback as a geometric allocation)**.** *Fix $k \in [0,\alpha]$, set $\lambda_k \triangleq \alpha - k$, and define*

$$q_i(k) = k\lambda_k^{i-1}, \qquad i \geq 1, \tag{24}$$

where $\lambda_k^0 \triangleq 1$. Then $\mathbf{q}(k) \in \mathcal{Q}_\alpha$ and $\pi_{\mathbf{q}(k)}$ is exactly the feasible proportional state-feedback policy $u_t(k) = kx_t(k)$. Its SDAC truncation is

$$\mathbf{m}_H^{\mathrm{prop}}(k) \triangleq \mathbf{m}_H\big(\mathbf{q}(k)\big),$$

with coordinates

$$m_{H,i}\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^{i-1}}{\alpha^i}, \qquad i \in [H],$$

and its tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Therefore, (23) *specializes to*

$$\begin{aligned}\left|\sum_{t=1}^{T}\left[c_t\big(x_t(\mathbf{m}_H^{\mathrm{prop}}(k)), u_t(\mathbf{m}_H^{\mathrm{prop}}(k))\big) - c_t\big(x_t(k), u_t(k)\big)\right]\right| \\ \leq \left(L_u + \frac{L_x}{1-\alpha}\right)\frac{k\lambda_k^H}{1-\lambda_k}\,T.\end{aligned} \tag{25}$$

Thus, $\Pi_\alpha^{\mathrm{alloc}}$ contains every feasible proportional state-feedback policy. Together with Theorem 1, the tail bound in (23) yields a no-regret guarantee against $\Pi_\alpha^{\mathrm{alloc}}$ and, consequently, against feasible proportional state-feedback policies.

IV. Online SDAC Controller

We now present our online SDAC controller. It updates its parameter over the subprobability simplex $\mathcal{M}_H \subset \mathbb{R}^H$ using OMD with unnormalized negentropy (equivalently, the exponentiated-gradient method) and a fixed step size $\eta > 0$ [21]. Let

$$\Psi(\mathbf{m}) \triangleq \sum_{i=1}^{H}\big(m_i \log m_i - m_i\big), \qquad \mathbf{m} \in \mathbb{R}_{\geq 0}^H, \tag{26}$$

denote the unnormalized negentropy (with the convention $0\log 0 \triangleq 0$). Its Bregman divergence is the generalized Kullback–Leibler divergence

$$D_\Psi(\mathbf{m}\|\mathbf{v}) \triangleq \sum_{i=1}^{H}\Big(m_i \log \tfrac{m_i}{v_i} - m_i + v_i\Big), \tag{27}$$

defined for

$$\mathbf{m} \in \mathbb{R}_{\geq 0}^H, \qquad \mathbf{v} \in \mathbb{R}_{>0}^H.$$

For each time $t \in [T]$, let $\mathbf{m}_t \in \mathcal{M}_H$ be the online parameter. We initialize at

$$\mathbf{m}_1 \triangleq \frac{1}{H+1}\mathbf{1} \tag{28}$$

so that, by (89), $\sup_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \log(H+1)$.

a) Control-action selection: At the start of round t, the controller has the current online parameter $\mathbf{m}_t$. For $t = 1$, it is given by the initialization (28), and for $t \geq 2$, it is the output of the OMD update (31) from round $t-1$. The controller then forms the nominal SDAC action

$$\widehat{u}_t \triangleq \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \tag{29}$$

and clips it to the admissible interval in (2):

$$u_t \triangleq \min\{\widehat{u}_t,\, \alpha x_t\}. \tag{30}$$

The state then evolves according to (3). Because $\widehat{u}_t \geq 0$, the clipping rule and induction on (3), starting from (1) with $w_t \in [0,1]$, give $x_t \geq 0$ for all $t \in [T+1]$. Thus, the online trajectory satisfies (2).

The clipping step also has a geometric interpretation. We use $\mathbf{m}_t^{\mathrm{exec}}$ to denote the execution vector obtained by projecting $\mathbf{m}_t$ onto the parameters compatible with the current state x_t, using the negentropy Bregman divergence. Lemma 6 formalizes this projection and shows that $\mathbf{m}_t^{\mathrm{exec}}$ induces exactly the clipped action u_t.

b) Surrogate cost and OMD update: After observing (w_t, c_t), choose $\mathbf{g}_t \in \partial\ell_t(\mathbf{m}_t)$. We define the OMD update

$$\mathbf{m}_{t+1} \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H}\left\{\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\right\}. \tag{31}$$

By Lemma 8, the minimizer in (31) is obtained in two steps. First, apply the multiplicative step,

$$\widetilde{m}_{t+1,i} \triangleq m_{t,i}\exp\big(-\eta g_{t,i}\big), \qquad i \in [H], \tag{32}$$

where $g_{t,i}$ is the i-th coordinate of $\mathbf{g}_t$. Next, apply the Bregman projection onto $\mathcal{M}_H$:

$$\mathbf{m}_{t+1} \triangleq \mathrm{Proj}_{\mathcal{M}_H}^{\Psi}(\widetilde{\mathbf{m}}_{t+1}) = \frac{\widetilde{\mathbf{m}}_{t+1}}{\max\{1, \|\widetilde{\mathbf{m}}_{t+1}\|_1\}}, \tag{33}$$

where Lemma 7 derives the projection formula.

V. Regret Upper Bounds

The following theorem states that the online SDAC controller achieves sublinear regret relative to every fixed SDAC policy $\pi_{\mathbf{m}} \in \Pi_H^{\mathrm{SDAC}}$. Proofs omitted from this section are collected in Appendix C.

Theorem 1 (Regret against fixed SDAC policies)**.** *Suppose the cost functions satisfy the regularity and bounded-subgradient assumptions of Section II. Then the controller from Section IV, initialized as in* (28) *and using step size $\eta > 0$, satisfies, for the regret defined in* (7)*,*

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le \frac{\log(H+1)}{\eta} + \eta T \Gamma_\alpha, \tag{34}$$

where

$$\Gamma_\alpha \triangleq \frac{G_\alpha^2}{2} + \frac{\alpha G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}, \tag{35}$$

and $G_\alpha = \alpha\big(\frac{L_x}{1-\alpha} + L_u\big)$ as in (17)*. In particular, the choice*

$$\eta^* = \sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}$$

gives

$$\mathrm{Reg}_T^{\mathrm{SDAC}} \le 2\sqrt{\Gamma_\alpha T \log(H+1)}.$$

For fixed $L_x, L_u > 0$, the leading factor obeys

$$\sqrt{\Gamma_\alpha} = \Theta(\alpha) \quad \text{as } \alpha \downarrow 0,$$

and

$$\sqrt{\Gamma_\alpha} = \Theta\Big((1-\alpha)^{-3/2}\Big) \quad \text{as } \alpha \uparrow 1.$$

The nonstandard step is to control the difference between the realized online trajectory and the fixed-policy trajectory associated with the current online parameter $\mathbf{m}_t$.

Lemma 5 (Stability under online parameter updates)**.** *For the online trajectory and the fixed-policy trajectory associated with the current online parameter $\mathbf{m}_t$, one has, for every $t \in [T]$,*

$$|x_t - x_t(\mathbf{m}_t)| \le \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \tag{36}$$

$$0 \le u_t(\mathbf{m}_t) - u_t \le \frac{\alpha^2 \eta G_\alpha}{(1-\alpha)^2}. \tag{37}$$

By definition of the fixed-policy trajectory, $x_t(\mathbf{m}_t)$ is generated by holding $\mathbf{m}_t$ constant from time 1 through time t. Consequently,

$$\sum_{t=1}^{T} \Big(c_t(x_t, u_t) - \ell_t(\mathbf{m}_t) \Big) \le \frac{\alpha\eta T G_\alpha (L_x + \alpha L_u)}{(1-\alpha)^2}. \tag{38}$$

The proof of Theorem 1 has three steps. First, the OMD bound for the surrogate costs gives

$$\sum_{t=1}^{T} \ell_t(\mathbf{m}_t) - \min_{\mathbf{m} \in \mathcal{M}_H} \sum_{t=1}^{T} \ell_t(\mathbf{m}) \le \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}.$$

Second, (38) controls the difference between the realized and surrogate costs. Third, adding the two bounds and using (35) gives (34); optimizing over η gives the stated choice η^*.

Theorem 1 and Lemma 4 give the following comparator guarantee.

Corollary 2 (No-regret against decreasing-allocation policies)**.** *Set*

$$H_T \triangleq \max\left\{1, \left\lceil \frac{\log T}{\log(1/\alpha)} \right\rceil \right\},$$

and run the online SDAC controller with action clipping, memory horizon $H = H_T$, and the step size η^* of Theorem 1. For $\mathbf{q} \in \mathcal{Q}_\alpha$, let

$$J_T(\mathbf{q}) \triangleq \sum_{t=1}^{T} c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)$$

and

$$\mathrm{Reg}_T^{\mathrm{alloc}} \triangleq J_T^{\mathrm{on}} - \inf_{\mathbf{q} \in \mathcal{Q}_\alpha} J_T(\mathbf{q}).$$

Then

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{alloc}} &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha} \right) \alpha^{H_T+1} T. \end{aligned} \tag{39}$$

Here Γ_α is defined in (35)*. Since $\alpha^{H_T} \le 1/T$ and $\log(H_T + 1) = O(\log\log T)$ as $T \to \infty$, the approximation term is $O(1)$. Consequently,*

$$\mathrm{Reg}_T^{\mathrm{alloc}} = O\Big(\sqrt{T \log\log T}\Big) = o(T).$$

Because geometric allocation sequences induce proportional state feedback, the controller also satisfies a gain-dependent bound for that policy family.

Corollary 3 (No-regret against proportional state-feedback policies)**.** *Run the online SDAC controller in Corollary 2. For $k \in [0, \alpha]$, let $J_T(k) \triangleq \sum_{t=1}^{T} c_t\big(x_t(k), u_t(k)\big)$, where $u_t(k) = k x_t(k)$, and define*

$$\mathrm{Reg}_T^{\mathrm{prop}} \triangleq J_T^{\mathrm{on}} - \min_{k \in [0,\alpha]} J_T(k).$$

For every $k \in [0, \alpha]$, with $\lambda_k = \alpha - k$,

$$\begin{aligned} J_T^{\mathrm{on}} - J_T(k) &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha} \right) \frac{k \lambda_k^{H_T}}{1-\lambda_k} T. \end{aligned} \tag{40}$$

Moreover,

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{prop}} &\le 2\sqrt{\Gamma_\alpha T \log(H_T+1)} \\ &\quad + \left(L_u + \frac{L_x}{1-\alpha} \right) \alpha^{H_T+1} T \\ &= O\Big(\sqrt{T \log\log T}\Big) = o(T). \end{aligned}$$

VI. Minimax Regret Lower Bound

The next result shows that the $\sqrt{T}$ dependence is unavoidable and quantifies the necessary dependence on $1-\alpha$ for any causal feasible controller. Proofs omitted from this section are collected in Appendix D.

Theorem 2 (Minimax SDAC regret lower bound)**.** *Fix $H \ge 1$ and $T \ge 2$. For every causal feasible online controller $\mathcal{A}$, possibly randomized, there exists a deterministic oblivious*

sequence of arrivals and costs satisfying the assumptions of Section II such that

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] \geq \frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\beta_t^2}, \tag{41}$$

where

$$\beta_t \triangleq \alpha\left(L_u + \frac{L_x(1-\alpha^{t-2})}{1-\alpha}\right). \tag{42}$$

The expectation is only over the internal randomization of $\mathcal{A}$; the adversarial sequence is deterministic and fixed before the interaction.

Corollary 4 (Lower bound beyond the mixing time)**.** *Define*

$$t_{\mathrm{mix}}(\alpha) \triangleq \left\lceil \frac{\log 2}{\log(1/\alpha)} \right\rceil. \tag{43}$$

If $T \geq 2t_{\mathrm{mix}}(\alpha)+2$, then every causal feasible controller $\mathcal{A}$ admits a deterministic oblivious sequence for which

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] \geq \frac{G_\alpha}{8}\sqrt{T}, \tag{44}$$

where $G_\alpha = \alpha(L_x/(1-\alpha)+L_u)$ as in (17)*.*

Remark 2 (Large-α regime). As $\alpha \uparrow 1$, stored arrivals persist longer and $t_{\mathrm{mix}}(\alpha) = \Theta((1-\alpha)^{-1})$. Hence, Corollary 4 yields

$$\mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] = \Omega\left(\frac{\sqrt{T}}{1-\alpha}\right) \tag{45}$$

whenever $T \geq 2t_{\mathrm{mix}}(\alpha)+2$. The upper bound in Theorem 1 is $O((1-\alpha)^{-3/2}\sqrt{T\log(H+1)})$, leaving a factor of $(1-\alpha)^{-1/2}$ between the two bounds.

The condition $T \geq 2t_{\mathrm{mix}}(\alpha)+2$ is essential for interpreting the large-α asymptotic. If T is fixed while $\alpha \uparrow 1$, then

$$\frac{1-\alpha^{t-2}}{1-\alpha} \longrightarrow t-2,$$

and the right-hand side of (41) converges to the finite quantity

$$\frac{1}{2\sqrt{2}}\sqrt{\sum_{t=2}^{T}\big(L_u + L_x(t-2)\big)^2}.$$

Thus, for fixed T, the exact finite-horizon bound in (41) applies instead of the long-horizon scaling in (45).

VII. Discussion and Extensions

We briefly discuss three extensions. In each case, the feasibility, convexity, and stability arguments used above continue to apply after the corresponding changes to the policy domain and the Lipschitz constants. We state these changes but omit the resulting regret bounds, whose form is unchanged up to the modified constants and, in the vector settings, the chosen product norm.

a) Retention $\alpha \geq 1$: For $\alpha \geq 1$, fix k such that $\bar{\alpha} \triangleq \alpha - k \in (0,1)$ and write $u_t = kx_t + v_t$. Then

$$x_{t+1} = \bar{\alpha}x_t - v_t + w_t.$$

An SDAC filter for v_t uses the powers $\bar{\alpha}^i$. If $0 \leq v_t \leq \bar{\alpha}x_t$, then

$$kx_t \leq u_t = kx_t + v_t \leq (k+\bar{\alpha})x_t = \alpha x_t,$$

so the original feasibility constraint (2) is satisfied. Since

$$x_{t+1} \leq \bar{\alpha}x_t + 1, \qquad x_1 = 0,$$

induction gives $x_t \leq 1/(1-\bar{\alpha})$. The transformed cost $c_t(x, kx+v)$ remains convex in (x,v). Thus the analysis applies when the costs satisfy the corresponding Lipschitz and bounded-subgradient assumptions on the bounded domain

$$0 \leq x \leq \frac{1}{1-\bar{\alpha}}, \qquad kx \leq u \leq \alpha x.$$

For a fixed k, the comparator class is correspondingly restricted to policies whose actions satisfy $u_t \geq kx_t$. Selecting k, either in advance or online, remains an interesting direction for future work.

b) Vector actions and multi-task allocation: Consider a single storage state x_t with a vector action $\mathbf{u}_t \in \mathbb{R}^n_{\geq 0}$ that allocates resources across n tasks, subject to $\sum_{a=1}^n u_{t,a} \leq \alpha x_t$. Thus, the total drain from storage is the sum of the task allocations. The dynamics remain $x_{t+1} = \alpha x_t - \sum_{a=1}^n u_{t,a} + w_t$, and the cost $c_t(x_t, \mathbf{u}_t)$ is convex in $(x, \mathbf{u})$. Replace $\mathbf{m} \in \mathcal{M}_H$ with a matrix $\mathbf{M} \in \mathbb{R}^{n\times H}_{\geq 0}$ satisfying

$$\sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1, \qquad u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} M_{a,i}\alpha^i w_{t-i}.$$

Set $m_i = \sum_{a=1}^n M_{a,i}$. Then

$$m_i \geq 0, \qquad \sum_{i=1}^{H} m_i = \sum_{a=1}^{n}\sum_{i=1}^{H} M_{a,i} \leq 1,$$

so $\mathbf{m} \in \mathcal{M}_H$. Moreover,

$$\sum_{a=1}^{n} u_{t,a}(\mathbf{M}) = \sum_{i=1}^{H} m_i\alpha^i w_{t-i} \leq \alpha x_t(\mathbf{M}),$$

where the inequality is the scalar SDAC feasibility bound. Both the state and vector action are affine in $\mathbf{M}$. Therefore,

$$c_t\big(x_t(\mathbf{M}), \mathbf{u}_t(\mathbf{M})\big)$$

is convex in $\mathbf{M}$. For the online SDAC controller, a nominal vector that exceeds the available total resource can be scaled proportionally so that its entries remain nonnegative and sum to αx_t. Under an ℓ_1 Lipschitz bound in the action, the size of this correction equals the excess total action, and the stability proof proceeds with the corresponding matrix norm. OMD then runs over the matrix subprobability simplex above.

c) Multiple storage states with diagonal dynamics: Suppose the state is $\mathbf{x}_t \in \mathbb{R}^d_{\geq 0}$ and evolves according to $\mathbf{x}_{t+1} = \mathbf{A}\mathbf{x}_t - \mathbf{u}_t + \mathbf{w}_t$, where $\mathbf{A} = \operatorname{diag}(\alpha_1, \dots, \alpha_d)$, $\alpha_s \in (0,1)$ for $s \in [d]$, the arrivals satisfy $\mathbf{w}_t \in [0,1]^d$, and the per-storage constraints are $0 \leq u_{t,s} \leq \alpha_s x_{t,s}$. Assign one SDAC parameter in $\mathcal{M}_H$ to each storage and use the powers α_s^i in coordinate s. The dynamics and feasibility constraints then decouple by storage, and the SDAC parameter domain is a product of subprobability simplices, with $x_{t,s} \leq 1/(1-\alpha_s)$ for every $s \in [d]$. The online SDAC controller clips each nominal action to $\alpha_s x_{t,s}$. If the cost is jointly convex and satisfies compatible Lipschitz and bounded-subgradient assumptions on the resulting bounded product domain, the surrogate cost is jointly convex in all SDAC parameters and OMD applies on the product set. The update separates by storage only when the cost is coordinate-separable; otherwise it is a joint update. Cross-coupled storage systems (nondiagonal $\mathbf{A}$) require additional analysis and remain an interesting direction for future work.

VIII. Conclusion

We studied adversarial online control of a storage system under a state-dependent action constraint. We introduced SDAC so that every fixed-policy trajectory is feasible and developed an online SDAC controller with $O\Big(\sqrt{T\log(H+1)}\Big)$ regret against the best fixed SDAC policy. Once T is at least on the order of $(1-\alpha)^{-1}$, the minimax lower bound gives $\Omega((1-\alpha)^{-1}\sqrt{T})$ regret as $\alpha \uparrow 1$; hence, a factor of $(1-\alpha)^{-1/2}$ between the upper and lower bounds remains open. The SDAC approximation results further give $O\big(\sqrt{T\log\log T}\big)$ regret against the best decreasing-allocation policy when $H = \Theta(\log T)$. The model applies directly to energy-harvesting battery management and related settings in which storage limits must be respected in real time.

Appendix A
Proofs for Simplex Disturbance-Action Control

Proof of Lemma 1. The auxiliary quantities satisfy

$$r_{t+1} = \alpha r_t + w_t, \qquad r_1 = 0, \tag{46}$$

and, for every $i \in \{1, \dots, H\}$,

$$\alpha r_t(i) - \alpha^i w_{t-i} = \sum_{j=1}^{i-1} \alpha^j w_{t-j} = r_{t+1}(i) - w_t. \tag{47}$$

For $i = 1$, the sum is empty.

At $t = 1$, since $w_\tau \equiv 0$ for $\tau \leq 0$, we have

$$r_1(i) = \sum_{j=1}^{i} \alpha^{j-1} w_{1-j} = 0$$

so (13) holds for $t = 1$. Suppose (13) holds at time t and plug it and (10) into dynamic equation (3) at $t+1$, we get

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \alpha\left(\sum_{i=1}^{H} m_i r_t(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_t\right) \\ &\quad - \sum_{i=1}^{H} m_i \alpha^i w_{t-i} + w_t \\ &= \sum_{i=1}^{H} m_i \big(\alpha r_t(i) - \alpha^i w_{t-i}\big) \\ &\quad + \alpha\big(1 - \sum_{i=1}^{H} m_i\big) r_t + w_t. \end{aligned} \tag{48}$$

Substituting (46) and (47) into (48),

$$\begin{aligned} x_{t+1}(\mathbf{m}) &= \sum_{i=1}^{H} m_i \big(r_{t+1}(i) - w_t\big) \\ &\quad + \big(1 - \sum_{i=1}^{H} m_i\big)(r_{t+1} - w_t) + w_t \\ &= \sum_{i=1}^{H} m_i r_{t+1}(i) + \big(1 - \sum_{i=1}^{H} m_i\big) r_{t+1}, \end{aligned} \tag{49}$$

The total coefficient of w_t in the first line is

$$-\sum_{i=1}^{H} m_i - \left(1 - \sum_{i=1}^{H} m_i\right) + 1 = 0.$$

Therefore, (13) follows by induction.

All summands defining $r_t(i)$ and r_t are nonnegative. Since $w_\tau = 0$ for $\tau \leq 0$, $r_t(i)$ is a truncation of r_t; hence

$$0 \leq r_t(i) \leq r_t \leq \sum_{j=1}^{\infty} \alpha^{j-1} = \frac{1}{1-\alpha}.$$

□

Proof of Lemma 3. Lemma 1 and (10) give

$$x_t(\mathbf{m}) = r_t + \langle \boldsymbol{\phi}_t^x, \mathbf{m} \rangle, \qquad u_t(\mathbf{m}) = \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle. \tag{50}$$

For every $i \in \{1, \dots, H\}$, the definitions of $r_t(i)$ and r_t give

$$\begin{aligned} \phi_{t,i}^x &= r_t(i) - r_t \\ &= -\sum_{j=i+1}^{t-1} \alpha^{j-1} w_{t-j}, \end{aligned} \tag{51}$$

where the sum is empty when $i \geq t-1$. Hence

$$\begin{aligned} \big|\phi_{t,i}^x\big| &\leq \sum_{j=i+1}^{\infty} \alpha^{j-1} = \frac{\alpha^i}{1-\alpha}, \\ 0 \leq \phi_{t,i}^u &= \alpha^i w_{t-i} \leq \alpha^i. \end{aligned} \tag{52}$$

Taking the maximum over the coordinates and noting that both upper bounds are largest at $i = 1$ yields

$$\|\boldsymbol{\phi}_t^x\|_\infty \leq \frac{\alpha}{1-\alpha}, \qquad \|\boldsymbol{\phi}_t^u\|_\infty \leq \alpha. \tag{53}$$

Since the state–action pair in (50) is affine in $\mathbf{m}$ and c_t is convex, ℓ_t is convex.

By (5) and (50),

$$\begin{aligned}|\ell_t(\mathbf{m}) - \ell_t(\mathbf{m}')| &\le L_x\big|\langle \boldsymbol{\phi}_t^x, \mathbf{m}-\mathbf{m}'\rangle\big| + L_u\big|\langle \boldsymbol{\phi}_t^u, \mathbf{m}-\mathbf{m}'\rangle\big| \\ &\le \big(L_x\|\boldsymbol{\phi}_t^x\|_\infty + L_u\|\boldsymbol{\phi}_t^u\|_\infty\big)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &\le \alpha\left(\frac{L_x}{1-\alpha} + L_u\right)\|\mathbf{m}-\mathbf{m}'\|_1 \\ &= G_\alpha\|\mathbf{m}-\mathbf{m}'\|_1. \end{aligned} \tag{54}$$

Let (ζ_t^x, ζ_t^u) denote the components of the bounded subgradient $\boldsymbol{\zeta}_t\big(x_t(\mathbf{m}), u_t(\mathbf{m})\big)$ from Section II, and set

$$\mathbf{g}_t \triangleq \zeta_t^x\, \boldsymbol{\phi}_t^x + \zeta_t^u\, \boldsymbol{\phi}_t^u. \tag{55}$$

For every $\mathbf{m}' \in \mathcal{M}_H$, the subgradient inequality and (50) yield

$$\begin{aligned}\ell_t(\mathbf{m}') &\ge \ell_t(\mathbf{m}) + \zeta_t^x\langle \boldsymbol{\phi}_t^x, \mathbf{m}'-\mathbf{m}\rangle + \zeta_t^u\langle \boldsymbol{\phi}_t^u, \mathbf{m}'-\mathbf{m}\rangle \\ &= \ell_t(\mathbf{m}) + \langle \mathbf{g}_t, \mathbf{m}'-\mathbf{m}\rangle.\end{aligned}$$

Thus $\mathbf{g}_t \in \partial\ell_t(\mathbf{m})$. For each $i \in \{1,\dots,H\}$, the exact coordinate decay in (52) gives

$$\begin{aligned}|g_{t,i}| &\le |\zeta_t^x|\big|\phi_{t,i}^x\big| + |\zeta_t^u|\phi_{t,i}^u \\ &\le \alpha^i\left(\frac{L_x}{1-\alpha} + L_u\right) \\ &\le \alpha\left(\frac{L_x}{1-\alpha} + L_u\right) = G_\alpha.\end{aligned} \tag{56}$$

Taking the maximum over i proves

$$\|\mathbf{g}_t\|_\infty \le G_\alpha. \tag{57}$$

□

Proof of Lemma 4. Fix $\mathbf{q} \in \mathcal{Q}_\alpha$ and set

$$a_i \triangleq \frac{q_i}{\alpha^i}, \qquad b_j \triangleq 1 - \sum_{i=1}^{j-1} a_i, \qquad j \ge 1.$$

The weighted-budget constraint gives $a_i \ge 0$ and $b_j \ge 0$. We first show that

$$x_t(\mathbf{q}) = \sum_{j=1}^{t-1} \alpha^{j-1} b_j w_{t-j}. \tag{58}$$

The identity is immediate at $t=1$. If it holds at time t, then

$$\begin{aligned}x_{t+1}(\mathbf{q}) &= w_t + \sum_{j=1}^{t-1}\big(\alpha^j b_j - q_j\big)w_{t-j} \\ &= w_t + \sum_{j=1}^{t-1}\alpha^j b_{j+1} w_{t-j},\end{aligned}$$

because $q_j = \alpha^j a_j$ and $b_{j+1} = b_j - a_j$. Reindexing gives (58) at time $t+1$.

For each $j \ge 1$,

$$a_j \le 1 - \sum_{i=1}^{j-1} a_i = b_j.$$

Consequently,

$$\begin{aligned}0 \le u_t(\mathbf{q}) &= \sum_{j=1}^{t-1}\alpha^j a_j w_{t-j} \\ &\le \sum_{j=1}^{t-1}\alpha^j b_j w_{t-j} = \alpha x_t(\mathbf{q}),\end{aligned}$$

which proves (2) for the decreasing-allocation policy.

For the truncated vector in (20),

$$m_{H,i}(\mathbf{q}) = a_i \ge 0, \qquad \sum_{i=1}^{H} m_{H,i}(\mathbf{q}) \le \sum_{i=1}^{\infty} a_i \le 1.$$

Thus, $\mathbf{m}_H(\mathbf{q}) \in \mathcal{M}_H$. Its SDAC action is

$$u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=1}^{\min\{H,t-1\}} q_i w_{t-i}.$$

It follows that

$$0 \le u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big) = \sum_{i=H+1}^{t-1} q_i w_{t-i} \le \mathrm{Tail}_H(\mathbf{q}),$$

where the middle sum is empty when $t \le H+1$. This proves (21).

Define

$$e_t^x \triangleq x_t\big(\mathbf{m}_H(\mathbf{q})\big) - x_t(\mathbf{q}), \qquad e_t^u \triangleq u_t(\mathbf{q}) - u_t\big(\mathbf{m}_H(\mathbf{q})\big).$$

The two state recursions give

$$e_{t+1}^x = \alpha e_t^x + e_t^u, \qquad e_1^x = 0.$$

Since $0 \le e_t^u \le \mathrm{Tail}_H(\mathbf{q})$,

$$0 \le e_t^x = \sum_{\tau=1}^{t-1}\alpha^{t-1-\tau} e_\tau^u \le \frac{\mathrm{Tail}_H(\mathbf{q})}{1-\alpha},$$

which proves (22). The Lipschitz property now yields

$$\begin{aligned}&\big|c_t\big(x_t(\mathbf{m}_H(\mathbf{q})), u_t(\mathbf{m}_H(\mathbf{q}))\big) \\ &\qquad - c_t\big(x_t(\mathbf{q}), u_t(\mathbf{q})\big)\big| \\ &\qquad \le \left(L_u + \frac{L_x}{1-\alpha}\right)\mathrm{Tail}_H(\mathbf{q}).\end{aligned}$$

Finally,

$$\begin{aligned}\mathrm{Tail}_H(\mathbf{q}) &= \sum_{i=H+1}^{\infty}\alpha^i a_i \\ &\le \alpha^{H+1}\sum_{i=H+1}^{\infty} a_i \le \alpha^{H+1}.\end{aligned}$$

Summing the per-period cost bound proves (23). □

Proof of Corollary 1. Fix $k \in [0,\alpha]$, let $\lambda_k = \alpha - k$, and let $\mathbf{q}(k)$ be defined by (24). The sequence is nonnegative and decreasing. If $k=0$, then $\mathbf{q}(k) = \mathbf{0} \in \mathcal{Q}_\alpha$. If $k > 0$, then $\lambda_k/\alpha < 1$ and

$$\sum_{i=1}^{\infty}\frac{q_i(k)}{\alpha^i} = \frac{k}{\alpha}\sum_{j=0}^{\infty}\left(\frac{\lambda_k}{\alpha}\right)^j = \frac{k}{\alpha-\lambda_k} = 1,$$

so again $\mathbf{q}(k) \in \mathcal{Q}_\alpha$. Substituting $u_t(k) = kx_t(k)$ into the dynamics and unrolling gives

$$u_t(k) = k\sum_{i=1}^{t-1} \lambda_k^{i-1} w_{t-i} = \sum_{i=1}^{t-1} q_i(k) w_{t-i} = u_t\big(\mathbf{q}(k)\big).$$

The two policies therefore have identical state and action trajectories. Their SDAC truncations also coincide coordinatewise, and the omitted geometric tail is

$$\mathrm{Tail}_H\big(\mathbf{q}(k)\big) = k\sum_{i=H+1}^{\infty} \lambda_k^{i-1} = \frac{k\lambda_k^H}{1-\lambda_k}.$$

Substitution in (23) proves (25). □

APPENDIX B
PROOFS FOR THE ONLINE SDAC CONTROLLER

The following lemmas establish the Bregman-projection interpretation of scalar action clipping, the projection formula, and the equivalent closed-form OMD update used in Section IV.

Lemma 6 (Action clipping as a Bregman projection)**.** *At time t, define the parameter set compatible with the available state by*

$$\mathcal{F}_t(x_t) \triangleq \{\mathbf{m} \in \mathcal{M}_H : \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \le \alpha x_t\}. \tag{59}$$

For the negentropy Ψ in (26)*, let*

$$\mathbf{m}_t^{\mathrm{exec}} \in \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{F}_t(x_t)} D_\Psi(\mathbf{m}\|\mathbf{m}_t). \tag{60}$$

Then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} = u_t. \tag{61}$$

Thus the projection formulation and the scalar clipping rule induce exactly the same executed action.

Proof. The set $\mathcal{F}_t(x_t)$ is nonempty because it contains $\mathbf{0}$, and it is compact and convex. By Lemma 8, every coordinate of $\mathbf{m}_t$ is strictly positive. Hence $D_\Psi(\cdot\|\mathbf{m}_t)$ is strictly convex on $\mathcal{M}_H$ and is minimized uniquely at $\mathbf{m}_t$ over any feasible set containing $\mathbf{m}_t$. Consequently, if

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle \le \alpha x_t,$$

then $\mathbf{m}_t^{\mathrm{exec}} = \mathbf{m}_t$, which proves (61) in the feasible case.

Suppose instead that $\mathbf{m}_t$ is infeasible. We claim that the action constraint is active at $\mathbf{m}_t^{\mathrm{exec}}$. If it were slack, then

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle < \alpha x_t < \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle. \tag{62}$$

For sufficiently small $\lambda \in (0,1)$, the point

$$\mathbf{m}_\lambda \triangleq (1-\lambda)\mathbf{m}_t^{\mathrm{exec}} + \lambda \mathbf{m}_t$$

would still belong to $\mathcal{F}_t(x_t)$. Because $D_\Psi(\mathbf{m}_t\|\mathbf{m}_t) = 0$, strict convexity would then give

$$\begin{aligned} D_\Psi(\mathbf{m}_\lambda\|\mathbf{m}_t) &< (1-\lambda) D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t) \\ &< D_\Psi(\mathbf{m}_t^{\mathrm{exec}}\|\mathbf{m}_t), \end{aligned}$$

contradicting (60). Hence

$$\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t^{\mathrm{exec}}\rangle = \alpha x_t$$

in the infeasible case, which completes the proof. □

Lemma 7 (Bregman projection onto $\mathcal{M}_H$)**.** *Let D_Ψ be the Bregman divergence* (27) *of the unnormalized negentropy* (26)*. Fix $\widetilde{\mathbf{m}} \in \mathbb{R}^H_{>0}$. Then the Bregman projection of $\widetilde{\mathbf{m}}$ onto $\mathcal{M}_H$,*

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) \triangleq \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}),$$

is given by

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\max\{1, \|\widetilde{\mathbf{m}}\|_1\}}.$$

Proof. The projection objective $\mathbf{m} \mapsto D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}})$ is strictly convex, and, for $m_i > 0$,

$$\frac{\partial}{\partial m_i} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}) = \log\frac{m_i}{\widetilde{m}_i}. \tag{63}$$

Its unconstrained minimizer is therefore $\mathbf{m} = \widetilde{\mathbf{m}}$.

If $\|\widetilde{\mathbf{m}}\|_1 \le 1$, this minimizer belongs to $\mathcal{M}_H$, and hence

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \widetilde{\mathbf{m}}.$$

Suppose $\|\widetilde{\mathbf{m}}\|_1 > 1$. The constraint $\|\mathbf{m}\|_1 \le 1$ is then active. Moreover, the minimizer has no zero coordinate: if $m_i = 0$, transferring an arbitrarily small amount of mass from a positive coordinate to coordinate i has directional derivative $-\infty$, because

$$\lim_{z\downarrow 0} \log\frac{z}{\widetilde{m}_i} = -\infty.$$

Thus the nonnegativity constraints are inactive. With $\nu \ge 0$ denoting the multiplier of $\|\mathbf{m}\|_1 \le 1$, stationarity and complementary slackness give

$$\log\frac{m_i}{\widetilde{m}_i} + \nu = 0, \qquad \|\mathbf{m}\|_1 = 1. \tag{64}$$

The first relation in (64) implies

$$m_i = \widetilde{m}_i \exp(-\nu).$$

Summing over i and using the second relation in (64),

$$1 = \|\widetilde{\mathbf{m}}\|_1 \exp(-\nu), \qquad \exp(-\nu) = \frac{1}{\|\widetilde{\mathbf{m}}\|_1}.$$

Hence $\nu = \log\|\widetilde{\mathbf{m}}\|_1 > 0$ and

$$\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}) = \frac{\widetilde{\mathbf{m}}}{\|\widetilde{\mathbf{m}}\|_1}.$$

Combining the cases $\|\widetilde{\mathbf{m}}\|_1 \le 1$ and $\|\widetilde{\mathbf{m}}\|_1 > 1$ proves the stated formula. □

Lemma 8 (Closed-form solution of the OMD update)**.** *Fix $\mathbf{m}_t \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$, $\eta > 0$, and $\mathbf{g}_t \in \mathbb{R}^H$, and let $\widetilde{\mathbf{m}}_{t+1}$ be defined coordinatewise by* (32)*, i.e. $\widetilde{m}_{t+1,i} = m_{t,i}\exp(-\eta g_{t,i})$. Then*

$$\begin{aligned} &\operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} \{\eta\langle \mathbf{g}_t, \mathbf{m}\rangle + D_\Psi(\mathbf{m}\|\mathbf{m}_t)\} \\ &\quad = \operatorname*{arg\,min}_{\mathbf{m}\in\mathcal{M}_H} D_\Psi(\mathbf{m}\|\widetilde{\mathbf{m}}_{t+1}) = \mathrm{Proj}^{\Psi}_{\mathcal{M}_H}(\widetilde{\mathbf{m}}_{t+1}), \end{aligned}$$

where $\mathrm{Proj}^{\Psi}_{\mathcal{M}_H}$ is the Bregman projection in Lemma 7. Consequently, the OMD update (31) *coincides exactly with the closed-form OMD update* (32)–(33)*, and, in particular,*

$\mathbf{m}_{t+1} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$. *Thus, the positive initialization* (28) *keeps every OMD iterate strictly positive.*

Proof. Fix $\mathbf{m} \in \mathbb{R}^H_{\geq 0}$ and a coordinate i. Then

$$\begin{aligned} \eta g_{t,i}\, m_i + m_i \log \frac{m_i}{m_{t,i}} &= m_i \left(\eta g_{t,i} + \log \frac{m_i}{m_{t,i}} \right) \\ &= m_i \log \frac{m_i}{\widetilde{m}_{t+1,i}}. \end{aligned} \tag{65}$$

Substitution into the OMD objective gives

$$\begin{aligned} &\eta \langle \mathbf{g}_t, \mathbf{m} \rangle + D_\Psi(\mathbf{m} \| \mathbf{m}_t) \\ &\qquad = D_\Psi(\mathbf{m} \| \widetilde{\mathbf{m}}_{t+1}) + \sum_{i=1}^H \left(m_{t,i} - \widetilde{m}_{t+1,i} \right). \end{aligned} \tag{66}$$

The final sum is independent of $\mathbf{m}$. Hence, the two optimization problems have the same minimizer, and Lemma 7 gives (33). Since $\mathbf{m}_t > 0$ coordinatewise, both the multiplicative step and the subsequent positive rescaling preserve strict positivity. □

APPENDIX C
PROOFS FOR THE REGRET UPPER BOUNDS

The following lemmas establish the OMD and stability estimates used in the proof of Theorem 1. They rely on the state-decomposition, feasibility, and surrogate-cost lemmas established in Section III (Lemmas 1, 2, and 3).

Lemma 9 (One-step OMD movement bound)**.** *For the OMD update with step size $\eta > 0$ and subgradient $\mathbf{g}_t \in \partial \ell_t(\mathbf{m}_t)$,*

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta \, \|\mathbf{g}_t\|_\infty.$$

In particular, since $\|\mathbf{g}_t\|_\infty \leq G_\alpha$, $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 \leq \eta G_\alpha$.

Proof. First-order optimality of (31), evaluated at the feasible point $\mathbf{m}_t$, gives

$$\begin{aligned} &\langle \eta \mathbf{g}_t + \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \geq 0, \\ &D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t) + D_\Psi(\mathbf{m}_t \| \mathbf{m}_{t+1}) \\ &\qquad \leq \eta \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\qquad \leq \eta \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{67}$$

The second line uses the symmetric Bregman identity.

For every $\mathbf{z} \in \mathcal{M}_H \cap \mathbb{R}^H_{>0}$ and $\mathbf{v} \in \mathbb{R}^H$,

$$\begin{aligned} \mathbf{v}^\top \nabla^2 \Psi(\mathbf{z}) \mathbf{v} &= \sum_{i=1}^H \frac{v_i^2}{z_i} \\ &\geq \frac{\|\mathbf{v}\|_1^2}{\sum_{i=1}^H z_i} \\ &\geq \|\mathbf{v}\|_1^2. \end{aligned} \tag{68}$$

The first inequality follows from Cauchy–Schwarz. The segment joining $\mathbf{m}_t$ and $\mathbf{m}_{t+1}$ lies in $\mathcal{M}_H \cap \mathbb{R}^H_{>0}$ by Lemma 8; integrating (68) along this segment yields

$$\begin{aligned} D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t) &\geq \frac{1}{2} \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2, \\ D_\Psi(\mathbf{m}_t \| \mathbf{m}_{t+1}) &\geq \frac{1}{2} \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2. \end{aligned}$$

Combining these inequalities with (67),

$$\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2 \leq \eta \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1.$$

The claim is immediate if $\mathbf{m}_{t+1} = \mathbf{m}_t$; otherwise divide by $\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1$. □

Lemma 10 (OMD regret for the surrogate costs)**.** *For every $\mathbf{m} \in \mathcal{M}_H$, the update* (31) *satisfies*

$$\sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) \leq \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_1)}{\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \tag{69}$$

Proof. Convexity of ℓ_t gives

$$\begin{aligned} \ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m} \rangle \\ &= \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle. \end{aligned} \tag{70}$$

First-order optimality of (31), evaluated at $\mathbf{m} \in \mathcal{M}_H$, yields

$$\begin{aligned} \eta \langle \mathbf{g}_t, \mathbf{m}_{t+1} - \mathbf{m} \rangle &\leq \langle \nabla\Psi(\mathbf{m}_{t+1}) - \nabla\Psi(\mathbf{m}_t), \mathbf{m} - \mathbf{m}_{t+1} \rangle \\ &= D_\Psi(\mathbf{m} \| \mathbf{m}_t) - D_\Psi(\mathbf{m} \| \mathbf{m}_{t+1}) \\ &\quad - D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t). \end{aligned} \tag{71}$$

The equality follows from the three-point Bregman identity. Substituting (71) into (70),

$$\begin{aligned} \ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m}) &\leq \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_t)}{\eta} - \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_{t+1})}{\eta} \\ &\quad + \langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle \\ &\quad - \frac{D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t)}{\eta}. \end{aligned} \tag{72}$$

The strong-convexity calculation (68) and Hölder's inequality give

$$\begin{aligned} &\langle \mathbf{g}_t, \mathbf{m}_t - \mathbf{m}_{t+1} \rangle - \frac{D_\Psi(\mathbf{m}_{t+1} \| \mathbf{m}_t)}{\eta} \\ &\qquad \leq \|\mathbf{g}_t\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1 - \frac{\|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1^2}{2\eta} \\ &\qquad \leq \frac{\eta}{2} \|\mathbf{g}_t\|_\infty^2. \end{aligned} \tag{73}$$

The last inequality follows from Young's inequality. Combining (72) and (73) and summing over t gives

$$\begin{aligned} \sum_{t=1}^T \ell_t(\mathbf{m}_t) - \sum_{t=1}^T \ell_t(\mathbf{m}) &\leq \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_1)}{\eta} \\ &\quad - \frac{D_\Psi(\mathbf{m} \| \mathbf{m}_{T+1})}{\eta} \\ &\quad + \frac{\eta}{2} \sum_{t=1}^T \|\mathbf{g}_t\|_\infty^2. \end{aligned}$$

Since $D_\Psi(\mathbf{m} \| \mathbf{m}_{T+1}) \geq 0$, this proves (69). □

Lemma 11 (Positive-part representation of the dynamics)**.** *For $\mathbf{m} \in \mathcal{M}_H$, define*

$$F_t(x; \mathbf{m}) \triangleq [\alpha x - \langle \boldsymbol{\phi}_t^u, \mathbf{m} \rangle]_+ + w_t. \tag{74}$$

Then the online state and the state component of every fixed-policy trajectory satisfy

$$x_{t+1} = F_t(x_t; \mathbf{m}_t), \qquad x_{t+1}(\mathbf{m}) = F_t(x_t(\mathbf{m}); \mathbf{m}). \tag{75}$$

Moreover, for all $x, x' \geq 0$ and $\mathbf{m}, \mathbf{m}' \in \mathcal{M}_H$,

$$\begin{aligned} &|F_t(x;\mathbf{m}) - F_t(x';\mathbf{m}')| \\ &\quad \leq |\alpha(x-x') - \langle \boldsymbol{\phi}_t^u, \mathbf{m}-\mathbf{m}'\rangle| \\ &\quad \leq \alpha|x-x'| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m}-\mathbf{m}'\|_1. \end{aligned} \tag{76}$$

Proof. For the online trajectory, (30) implies

$$\begin{aligned} \alpha x_t - u_t &= \alpha x_t - \min\{\langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle, \alpha x_t\} \\ &= \left[\alpha x_t - \langle \boldsymbol{\phi}_t^u, \mathbf{m}_t\rangle\right]_+. \end{aligned} \tag{77}$$

For a fixed $\mathbf{m}$, Lemma 2 gives

$$\alpha x_t(\mathbf{m}) - \langle \boldsymbol{\phi}_t^u, \mathbf{m}\rangle \geq 0,$$

so inserting a positive part does not change its state update. This proves (75). Finally, the map $z \mapsto [z]_+$ is nonexpansive:

$$|[z]_+ - [z']_+| \leq |z - z'|.$$

Applying this inequality to the two arguments in (74), followed by Hölder's inequality, gives (76). □

Proof of Lemma 5. Define

$$d_t \triangleq |x_t - x_t(\mathbf{m}_t)|.$$

Since $x_1 = x_1(\mathbf{m}_1) = 0$, we have $d_1 = 0$. For $t \in \{1, \dots, T-1\}$, Lemma 11 gives

$$\begin{aligned} d_{t+1} &= |F_t(x_t;\mathbf{m}_t) - F_t(x_t(\mathbf{m}_{t+1});\mathbf{m}_{t+1})| \\ &\leq \alpha\,|x_t - x_t(\mathbf{m}_{t+1})| + \|\boldsymbol{\phi}_t^u\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{78}$$

The affine state representation (50) implies

$$\begin{aligned} |x_t(\mathbf{m}_t) - x_t(\mathbf{m}_{t+1})| &= |\langle \boldsymbol{\phi}_t^x, \mathbf{m}_t - \mathbf{m}_{t+1}\rangle| \\ &\leq \|\boldsymbol{\phi}_t^x\|_\infty \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{79}$$

Combining the triangle inequality with (78) and (79) yields

$$\begin{aligned} d_{t+1} &\leq \alpha d_t \\ &\quad + \left(\alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty\right) \|\mathbf{m}_{t+1} - \mathbf{m}_t\|_1. \end{aligned} \tag{80}$$

By (53),

$$\begin{aligned} \alpha\|\boldsymbol{\phi}_t^x\|_\infty + \|\boldsymbol{\phi}_t^u\|_\infty &\leq \frac{\alpha^2}{1-\alpha} + \alpha \\ &= \frac{\alpha}{1-\alpha}. \end{aligned} \tag{81}$$

Lemma 9 and $\|\mathbf{g}_t\|_\infty \leq G_\alpha$ therefore reduce (80) to

$$d_{t+1} \leq \alpha d_t + \frac{\alpha\eta G_\alpha}{1-\alpha}. \tag{82}$$

Unrolling from $d_1 = 0$ gives

$$\begin{aligned} d_t &\leq \frac{\alpha\eta G_\alpha}{1-\alpha} \sum_{j=0}^{t-2} \alpha^j \\ &\leq \frac{\alpha\eta G_\alpha}{(1-\alpha)^2}, \end{aligned} \tag{83}$$

with an empty sum when $t = 1$. This proves (36).

Finally, $\widehat{u}_t = u_t(\mathbf{m}_t)$ and the feasibility of fixed SDAC policies imply

$$\begin{aligned} 0 \leq u_t(\mathbf{m}_t) - u_t &= \left[u_t(\mathbf{m}_t) - \alpha x_t\right]_+ \\ &\leq \left[\alpha x_t(\mathbf{m}_t) - \alpha x_t\right]_+ \\ &\leq \alpha d_t. \end{aligned} \tag{84}$$

Combining (84) with (36) proves (37).

Both state–action pairs lie in $\mathcal{D}_\alpha$. Hence (5) and the two bounds just proved give

$$\begin{aligned} &c_t(x_t,u_t) - \ell_t(\mathbf{m}_t) \\ &\quad \leq L_x\big|x_t - x_t(\mathbf{m}_t)\big| + L_u\big|u_t - u_t(\mathbf{m}_t)\big| \\ &\quad \leq \frac{\alpha\eta G_\alpha L_x}{(1-\alpha)^2} + \frac{\alpha^2\eta G_\alpha L_u}{(1-\alpha)^2} \\ &\quad = \frac{\alpha\eta G_\alpha(L_x + \alpha L_u)}{(1-\alpha)^2}. \end{aligned} \tag{85}$$

Summing (85) over $t = 1, \dots, T$ proves (38). □

Proof of Theorem 1. Define the stability term

$$\mathrm{Stab}_T \triangleq \sum_{t=1}^{T} \left(c_t(x_t,u_t) - \ell_t(\mathbf{m}_t)\right).$$

Using $\ell_t(\mathbf{m}) = c_t(x_t(\mathbf{m}), u_t(\mathbf{m}))$, the regret decomposes as

$$\mathrm{Reg}_T^{\mathrm{SDAC}} = \mathrm{Stab}_T + \sum_{t=1}^{T} \ell_t(\mathbf{m}_t) - \min_{\mathbf{m}\in\mathcal{M}_H} \sum_{t=1}^{T} \ell_t(\mathbf{m}). \tag{86}$$

We first bound the initial Bregman divergence. Fix $\mathbf{m} \in \mathcal{M}_H$. Since $\mathbf{m}_1 = \frac{1}{H+1}\mathbf{1}$,

$$\begin{aligned} D_\Psi(\mathbf{m}\|\mathbf{m}_1) &= \sum_{i=1}^{H} m_i \log\big((H+1)m_i\big) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1} \\ &= \sum_{i=1}^{H} m_i \log m_i + \|\mathbf{m}\|_1 \log(H+1) \\ &\quad - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \end{aligned} \tag{87}$$

Because $z \log z \leq 0$ for $z \in [0,1]$,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \|\mathbf{m}\|_1 \log(H+1) - \|\mathbf{m}\|_1 + \frac{H}{H+1}. \tag{88}$$

The right-hand side is affine in $\|\mathbf{m}\|_1 \in [0,1]$, and its values at the endpoints are

$$\begin{aligned} \frac{H}{H+1} &\leq \log(H+1), \\ \log(H+1) - \frac{1}{H+1} &\leq \log(H+1). \end{aligned}$$

Here the first inequality follows from $\log z \geq 1 - z^{-1}$ for $z \geq 1$. Therefore,

$$D_\Psi(\mathbf{m}\|\mathbf{m}_1) \leq \log(H+1) \qquad \text{for every } \mathbf{m} \in \mathcal{M}_H. \tag{89}$$

Lemma 10, $\|\mathbf{g}_t\|_\infty \leq G_\alpha$, and (89) imply, for every $\mathbf{m} \in \mathcal{M}_H$,

$$\begin{aligned} \sum_{t=1}^{T} \big(\ell_t(\mathbf{m}_t) - \ell_t(\mathbf{m})\big) &\leq \frac{D_\Psi(\mathbf{m}\|\mathbf{m}_1)}{\eta} + \frac{\eta}{2}\sum_{t=1}^{T} \|\mathbf{g}_t\|_\infty^2 \\ &\leq \frac{\log(H+1)}{\eta} + \frac{\eta G_\alpha^2 T}{2}. \end{aligned} \tag{90}$$

Since (90) holds uniformly over $\mathbf{m}$,

$$\sum_{t=1}^{T}\ell_t(\mathbf{m}_t)-\min_{\mathbf{m}\in\mathcal{M}_H}\sum_{t=1}^{T}\ell_t(\mathbf{m})\le\frac{\log(H+1)}{\eta}+\frac{\eta G_\alpha^2 T}{2}. \tag{91}$$

The stability bound (38) gives

$$\mathrm{Stab}_T\le\frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}. \tag{92}$$

Substituting (91) and (92) into (86),

$$\begin{aligned}\mathrm{Reg}_T^{\mathrm{SDAC}}&\le\frac{\log(H+1)}{\eta}+\frac{\eta T G_\alpha^2}{2}\\&\quad+\frac{\alpha\eta T G_\alpha(L_x+\alpha L_u)}{(1-\alpha)^2}\\&=\frac{\log(H+1)}{\eta}+\eta T\Gamma_\alpha.\end{aligned} \tag{93}$$

Differentiating with respect to η yields

$$-\frac{\log(H+1)}{\eta^2}+T\Gamma_\alpha.$$

The unique minimizer is therefore

$$\eta^*=\sqrt{\frac{\log(H+1)}{T\Gamma_\alpha}}. \tag{94}$$

At this value, the two summands in (93) are equal, so

$$\mathrm{Reg}_T^{\mathrm{SDAC}}\le 2\sqrt{\Gamma_\alpha T\log(H+1)}. \tag{95}$$

It remains to verify the dependence on α. As $\alpha\downarrow 0$,

$$\begin{aligned}\frac{G_\alpha}{\alpha}&=\frac{L_x}{1-\alpha}+L_u\longrightarrow L_x+L_u,\\\frac{\Gamma_\alpha}{\alpha^2}&=\frac{1}{2}\left(\frac{G_\alpha}{\alpha}\right)^2+\frac{G_\alpha}{\alpha}\frac{L_x+\alpha L_u}{(1-\alpha)^2}\\&\longrightarrow\frac{(L_x+L_u)^2}{2}+L_x(L_x+L_u)>0.\end{aligned}$$

Therefore $\sqrt{\Gamma_\alpha}=\Theta(\alpha)$ as $\alpha\downarrow 0$. As $\alpha\uparrow 1$, write $\varepsilon=1-\alpha$. Then

$$\begin{aligned}\varepsilon G_\alpha&=\alpha(L_x+\varepsilon L_u)\longrightarrow L_x,\\\varepsilon^3\Gamma_\alpha&=\frac{\varepsilon^3 G_\alpha^2}{2}+\alpha(\varepsilon G_\alpha)(L_x+\alpha L_u)\\&\longrightarrow L_x(L_x+L_u)>0.\end{aligned}$$

Thus $\sqrt{\Gamma_\alpha}=\Theta((1-\alpha)^{-3/2})$ as $\alpha\uparrow 1$. □

Proof of Corollary 2. Fix $\mathbf{q}\in\mathcal{Q}_\alpha$ and use the SDAC comparator $\mathbf{m}_{H_T}(\mathbf{q})$ from Lemma 4. Theorem 1 and (23) give

$$\begin{aligned}J_T^{\mathrm{on}}-J_T(\mathbf{q})&\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\&\quad+\left(L_u+\frac{L_x}{1-\alpha}\right)\mathrm{Tail}_{H_T}(\mathbf{q})T\\&\le 2\sqrt{\Gamma_\alpha T\log(H_T+1)}\\&\quad+\left(L_u+\frac{L_x}{1-\alpha}\right)\alpha^{H_T+1}T.\end{aligned}$$

The bound is uniform in $\mathbf{q}$, so taking the infimum over $\mathbf{q}\in\mathcal{Q}_\alpha$ proves (39). Finally, $H_T\ge\log T/\log(1/\alpha)$ implies $\alpha^{H_T}\le 1/T$, while $H_T=\Theta(\log T)$ implies $\log(H_T+1)=O(\log\log T)$. □

Proof of Corollary 3. For fixed k, Theorem 1 applied to $\mathbf{m}_{H_T}^{\mathrm{prop}}(k)$, followed by (25), gives (40). Corollary 1 shows that the geometric allocation sequence $\mathbf{q}(k)$ belongs to $\mathcal{Q}_\alpha$, so Lemma 4 gives $\mathrm{Tail}_{H_T}(\mathbf{q}(k))\le\alpha^{H_T+1}$. This makes the bound uniform in k; minimizing over k proves the stated finite-time and asymptotic guarantees. □

APPENDIX D
PROOFS FOR THE MINIMAX REGRET LOWER BOUND

Proof of Theorem 2. Fix a causal feasible controller $\mathcal{A}$. Let $\xi_1,\dots,\xi_T$ be independent Rademacher random variables, also independent of the internal randomization of $\mathcal{A}$, with

$$\mathbb{P}(\xi_t=1)=\mathbb{P}(\xi_t=-1)=\frac{1}{2}.$$

Set

$$w_t=1\qquad\text{for all } t\in[T], \tag{96}$$

and define the costs by

$$c_t(x,u)=\xi_t\big(L_x x-L_u u\big)\qquad\text{for all } t\in[T]. \tag{97}$$

Each c_t is affine and hence convex. For any two points in $\mathcal{D}_\alpha$,

$$\begin{aligned}|c_t(x,u)-c_t(x',u')|&=|L_x(x-x')-L_u(u-u')|\\&\le L_x|x-x'|+L_u|u-u'|.\end{aligned} \tag{98}$$

Moreover,

$$\nabla c_t=\xi_t(L_x,-L_u),\qquad|\partial_x c_t|=L_x,\qquad|\partial_u c_t|=L_u,$$

so the required subgradient bounds hold.

We now select two fixed SDAC policies for the benchmark. The parameter $\mathbf{m}^{(0)}$ leaves all mass in the subprobability simplex unused, so its SDAC action is always zero. The parameter $\mathbf{m}^{(1)}$ places all mass on the first lag, so its action is αw_{t-1}. Let

$$\mathbf{m}^{(0)}\triangleq\mathbf{0},\qquad\mathbf{m}^{(1)}\triangleq\mathbf{e}_1 \tag{99}$$

belong to $\mathcal{M}_H$, where $\mathbf{e}_1=(1,0,\dots,0)\in\mathbb{R}^H$. Thus, under (96), $\mathbf{m}^{(0)}$ is the no-release policy, whereas $\mathbf{m}^{(1)}$ releases the retained portion of the preceding arrival. For the no-release policy,

$$u_t(\mathbf{m}^{(0)})=0,\qquad x_t(\mathbf{m}^{(0)})=\rho_t\triangleq\frac{1-\alpha^{t-1}}{1-\alpha}. \tag{100}$$

This follows by unrolling $x_{t+1}=\alpha x_t+1$:

$$x_t(\mathbf{m}^{(0)})=\sum_{s=0}^{t-2}\alpha^s=\frac{1-\alpha^{t-1}}{1-\alpha},\qquad t\ge 2,$$

and the closed form also gives $x_1(\mathbf{m}^{(0)})=0$. For the one-lag policy, (10) gives

$$u_1(\mathbf{m}^{(1)})=0,\qquad u_t(\mathbf{m}^{(1)})=\alpha,\quad t\ge 2. \tag{101}$$

Its state at $t=2$ is

$$x_2(\mathbf{m}^{(1)})=\alpha x_1(\mathbf{m}^{(1)})-u_1(\mathbf{m}^{(1)})+w_1=1.$$

If $x_t(\mathbf{m}^{(1)}) = 1$ for some $t \geq 2$, then

$$\begin{aligned} x_{t+1}(\mathbf{m}^{(1)}) &= \alpha x_t(\mathbf{m}^{(1)}) - u_t(\mathbf{m}^{(1)}) + w_t \\ &= \alpha - \alpha + 1 = 1. \end{aligned}$$

Induction therefore gives

$$x_t(\mathbf{m}^{(1)}) = 1, \qquad t \geq 2. \tag{102}$$

For $t \geq 2$, the difference between the two states is

$$\begin{aligned} \Delta_t &\triangleq x_t(\mathbf{m}^{(0)}) - x_t(\mathbf{m}^{(1)}) \\ &= \rho_t - 1 \\ &= \frac{\alpha(1-\alpha^{t-2})}{1-\alpha}. \end{aligned} \tag{103}$$

The two policies have the same state and action at $t = 1$. For $t \geq 2$, (101) and (103) give

$$\begin{aligned} J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)}) &= \sum_{t=2}^{T} \xi_t (L_x \Delta_t + \alpha L_u) \\ &= \sum_{t=2}^{T} \xi_t \beta_t, \end{aligned} \tag{104}$$

with β_t as in (42).

Let $\mathcal{F}_t^-$ be the information available immediately before c_t is revealed, including the randomization used to select u_t. By causality, (x_t, u_t) is $\mathcal{F}_t^-$-measurable, whereas ξ_t is independent of $\mathcal{F}_t^-$ and has mean zero. Hence, by the tower property,

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}[J_T^{\mathrm{on}}] &= \sum_{t=1}^{T} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\big(L_x x_t - L_u u_t\big)\mathbb{E}[\xi_t \mid \mathcal{F}_t^-]\big] \\ &= 0. \end{aligned} \tag{105}$$

The fixed-policy trajectories depend only on the arrival sequence and are therefore independent of the Rademacher variables. Thus, for $j \in \{0, 1\}$,

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi}}\Big[J_T(\mathbf{m}^{(j)})\Big] &= \sum_{t=1}^{T} \mathbb{E}[\xi_t]\big(L_x x_t(\mathbf{m}^{(j)}) - L_u u_t(\mathbf{m}^{(j)})\big) \\ &= 0. \end{aligned} \tag{106}$$

Because $\mathbf{m}^{(0)}, \mathbf{m}^{(1)} \in \mathcal{M}_H$, for every realization of the Rademacher variables and every choice made by the controller,

$$\begin{aligned} \mathrm{Reg}_T^{\mathrm{SDAC}} &= J_T^{\mathrm{on}} - \min_{\mathbf{m} \in \mathcal{M}_H} J_T(\mathbf{m}) \\ &\geq J_T^{\mathrm{on}} - \min\Big\{J_T(\mathbf{m}^{(0)}), J_T(\mathbf{m}^{(1)})\Big\}. \end{aligned} \tag{107}$$

Taking expectations in (107) and using $\min\{a, b\} = (a + b - |a - b|)/2$ together with (105) and (106) gives

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] &\geq \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\Big[\Big|J_T(\mathbf{m}^{(0)}) - J_T(\mathbf{m}^{(1)})\Big|\Big] \\ &= \frac{1}{2}\mathbb{E}_{\boldsymbol{\xi}}\left[\left|\sum_{t=2}^{T} \xi_t \beta_t\right|\right] \\ &\geq \frac{1}{2\sqrt{2}}\left(\sum_{t=2}^{T} \beta_t^2\right)^{1/2}. \end{aligned} \tag{108}$$

Khintchine's inequality for Rademacher sums gives the final inequality in (108). It remains to remove the randomization from the adversarial construction. The left-hand side of (108) is the average over all 2^T realizations of the Rademacher vector $\boldsymbol{\xi}$:

$$\mathbb{E}_{\boldsymbol{\xi},\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] = \frac{1}{2^T} \sum_{\boldsymbol{\xi} \in \{-1,1\}^T} \mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}} \,\big|\, \boldsymbol{\xi}\big].$$

Hence, at least one vector $\boldsymbol{\xi}^\star$ makes the conditional expectation at least the right-hand side of (108). Fixing $\boldsymbol{\xi}^\star$ in (97), together with the arrivals in (96), produces a deterministic admissible sequence fixed before the interaction. The remaining expectation is only over the internal randomization of $\mathcal{A}$, and (41) follows. □

Proof of Corollary 4. The definition of $t_{\mathrm{mix}}(\alpha)$ gives

$$\alpha^{t_{\mathrm{mix}}(\alpha)} \leq \alpha^{\log 2/\log(1/\alpha)} = \frac{1}{2}. \tag{109}$$

Therefore, for every $t \geq t_{\mathrm{mix}}(\alpha) + 2$,

$$\begin{aligned} \beta_t &\geq \alpha\left(L_u + \frac{L_x}{2(1-\alpha)}\right) \\ &\geq \frac{\alpha}{2}\left(L_u + \frac{L_x}{1-\alpha}\right) = \frac{G_\alpha}{2}. \end{aligned} \tag{110}$$

There are $T - t_{\mathrm{mix}}(\alpha) - 1$ indices in $\{t_{\mathrm{mix}}(\alpha) + 2, \ldots, T\}$. If $T \geq 2t_{\mathrm{mix}}(\alpha) + 2$, then

$$T - t_{\mathrm{mix}}(\alpha) - 1 \geq \frac{T}{2}. \tag{111}$$

Applying Theorem 2 and retaining only these indices,

$$\begin{aligned} \mathbb{E}_{\mathcal{A}}\big[\mathrm{Reg}_T^{\mathrm{SDAC}}\big] &\geq \frac{1}{2\sqrt{2}}\left[\frac{T}{2}\left(\frac{G_\alpha}{2}\right)^2\right]^{1/2} \\ &= \frac{G_\alpha}{8}\sqrt{T}. \end{aligned} \tag{112}$$

This is (44). □

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